Digital Twin-Driven Deep RL for Energy-Aware CNC Milling

Dynamic Cutting Parameter Optimization

Problem Statement

Challenges in CNC Machining:

- Optimal cutting parameters (speed, feed, depth) are critical to balance throughput and quality.
- Traditional meta-heuristic algorithms optimize these statically.
- Real-world multi-pass operations experience dynamic tool degradation (wear and stochastic breakage).

Modern Manufacturing Needs:

- Requires energy-efficient operations.
- Static approaches cannot adapt to rapidly changing tool conditions on the fly.

Proposed Methodology : Deep RL

MDP Formulation

$$S_t = (VB_i, \kappa, n_{i-1}, f_{i-1}, ap_{i-1}, MRR_{i-1}, Ra_{i-1}, B_i) \in [0,1]^8$$

- VB_i (Tool Wear): The current physical flank wear of the cutting tool (measured in mm).
- κ (Remaining Allowance): The total depth of material that still needs to be removed to finish the workpiece (mm)
- n_{i-1} (Previous Spindle Speed): The rotational speed of the spindle used during the last cutting pass (RPM).
- f_{i-1} (Previous Feed Rate): The speed at which the tool advanced against the workpiece during the last pass (mm/rev)
- ap_{i-1} (Previous Depth of Cut): The depth of the cut made during the previous pass (measured in mm).
- MRR_{i-1} (Previous Material Removal Rate): The volume of material removed per minute during the last pass, indicating cutting throughput (measured in mm^3/min).
- Ra_{i-1} (Previous Surface Roughness): The calculated surface finish quality achieved on the workpiece after the last pass (measured in µm).
- B_i (Tool Breakage State): A boolean safety flag (1 if the tool has catastrophically broken, 0 if it is still intact).

Action Space

The agent outputs a continuous 3-dimensional vector defining the cutting parameters for the next pass.

$$A_t = (n, f_r, a_p)$$

- n (Spindle Speed): The rotational speed of the cutting tool (Bounded: 2000 to 8000 RPM).
- f_r (Feed Rate): The distance the tool advances per revolution (Bounded: 0.05 to 0.35 mm/r).
- a_p (Depth of Cut): The depth the tool plunges into the material (Bounded: 0.1 to 2.5 mm).

Reward Function

$$\mathbf{r_t = r_{VB} + r_B + r_{Ra} + r_{MRR} + r_{Energy} + r_{HV} + r_{step}}$$

$$r_{VB} = \begin{cases} 0, & VB \leq VB_{max} \\ -50, & \text{otherwise} \end{cases}$$

$$r_B = \begin{cases} 0, & \text{Tool intact} \\ -100, & \text{Tool broken} \end{cases}$$

$$r_{Ra} = \begin{cases} 0, & Ra \leq Ra_{max} \\ -50, & \text{otherwise} \end{cases}$$

$$r_{MRR} = 0.5 \left(\frac{MRR}{MRR_{max}} \right)$$

$$r_{Energy} = -w_2 \left(\frac{E}{E_{max}} \right)$$

$$r_{HV} = \left(\frac{\max(0, T_{ref} - T_{total})}{T_{ref}} \right) \times \left(\frac{\max(0, Ra_{ref} - Ra)}{Ra_{ref}} \right) \times 100$$

$$r_{step} = -0.1$$

Reinforcement Learning Algorithm

The chosen algorithm is **Proximal Policy Optimization (PPO)**.

- PPO is an on-policy algorithm that strikes an excellent balance between sample efficiency and ease of tuning.
- Given the continuous nature of the action space and the complexity of the multi-objective delayed Hypervolume rewards.
- PPO ensures monotonic policy improvements and stable gradient updates, which are essential for converging toward a robust, constraint-compliant machining policy while mitigating the risk of Knowledge degradation and Unbounded policy divergence